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COLLEGE OF NATURAL SCIENCE

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Final Year Project Proposal

Project Title: Online Job Portal System in Ethiopia

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Acronyms

CV: Curriculum Vitae

HTTP: Hypertext Transfer Protocol

OOM: Object-Oriented Modeling

OJPS: Online Job Portal System

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1. Introduction

In an ever-evolving global job market, the need for a dynamic and efficient platform connecting employers with talented individuals is more crucial than ever. An online job portal is a website that enables job seekers and employers to connect and communicate in an easy and efficient way [1]. However, having an online job portal is not that common in countries like Ethiopia, mostly those higher-class businesses and organizations engage themselves in this kind of scenario, but with the growth in economy and technology, more people will be able to access and benefit from online job portals. In most areas across Ethiopia, the predominant use of job portals is by educated individuals and higher-class businesses and organizations. Traditional methods such as newspapers, word-of-mouth, or personal contacts are still used to find and offer jobs. This indicates dissatisfaction with the current job portal systems, which may not be accessible or affordable for some users. Our system aims to bridge this gap by providing a feature-rich environment for job search and recruitment. This project aims to explore the incentives and barriers to creating and adopting online job portals in Ethiopia. The online job portal project is also expected to have a positive impact on the academic and professional development of the project team, as it will enhance our skills and knowledge in web development.

1.1. Background Information

Traditional job-seeking methods are becoming outdated, and there is a growing demand for a streamlined and accessible online solution. Recognizing this paradigm shift, our proposal introduces an innovative online job portal designed to revolutionize the way job seekers and employers connect. This digital platform aims to address the challenges faced by both job seekers and employers.

2. The existing System

Currently, the process of discovering job opportunities involves both traditional methods and online platforms.

- **Traditional Methods**

Traditional methods, such as magazines, posters, physical job fairs, and board advertisements, have been the longstanding avenues for job discovery. These methods offer familiarity,

accessibility, and affordability. Job seekers can easily access information on job openings in physical spaces, making these methods inclusive for a wide range of individuals.

- **Online Methods**

Online job portals and social media platforms have become integral components of the modern job search process. Online job portals, in particular, have streamlined the job search process by centralizing job listings and providing search functionalities, most of the job portal platforms operate at some cost. In recent years, social media platforms have emerged as dynamic hubs for job seekers to explore diverse job opportunities. Among these, Telegram and other social media channels play a pivotal role in connecting individuals with suitable job openings.

3. Statement of the problem

Many individuals from diverse backgrounds struggle to secure stable employment, even the first challenge for a post-graduate is finding a suitable job to kickstart their career, while other members of our society struggle to secure work that can sustain them financially and on the other hand Employers often encounter difficulties in quickly and effectively identifying the right candidates for their open positions. here are some current problems in the existing system

- Outdated and unreliable for real-time job updates while using traditional methods.
- Limited in scope, hindering a comprehensive job search.
- The paid subscription model creates financial barriers for job seekers while using job portals.
- Decentralized nature may result in a fragmented job search experience in social media.
- The absence of supportive features, such as intelligent assistance.
- Lack of structure in the job search process on some platforms such as Telegram.

4. Objective of the Study

4.1. General Objectives

To develop an online job portal system in Ethiopia that provides a connection between job seekers and employers and eases the effort for users' job searching experience.

4.2. Specific Objectives

In order to achieve the major objective of the proposed project, the following specific objectives are set.

- ✓ To Gather requirements from the existing system that are relevant to the project.
- ✓ To Analyze the requirements of the new system that is to be developed.
- ✓ To Design the system based on the specified requirements in the analysis phase, by OOM.
- ✓ Implementing the system based on the design.
- ✓ Testing the system to verify that all the required functionalities are implemented.
- ✓ To deploy the system in the working environment.

5. Scope of the project

The online job portal project encompasses a transformative vision to redefine employment connections in the Ethiopian job market.

- Employers can post job listings, access candidate, and view applicants.
- Job seekers can create personalized profiles, apply for posted jobs, and download their CVs based on their profiles.
- The platform will have user-friendly documentation and mobile responsiveness.
- It will be integrated with external platforms.

6. Significance of the project

Online job portals are an essential tool for both job seekers and employers. the project lies in its potential to bring about positive and impactful changes to the employment landscape in the Ethiopian job market. Here are the key aspects of its significance:

- **Saves time and money**: The job portal saves time and money for both job seekers and employers.
- **Wider reach**: Have a wider reach than traditional methods of job search.
- **Efficient filtering**: Use search feature to match job seekers with relevant job openings based on their search.
- **Easy to use**: easy to use and navigate.
- **Download CV**: used to download a CV based on the profile.

No	Actors	Benefits of the project
1	**Job seeker**	✓ Easy job application process ✓ Document CV ✓ Access to an assistance ✓ User-friendly system ✓ Better visibility to employers ✓ Access to a wide range of job opportunities
2	**Employer**	✓ Time and cost efficiency ✓ Enhanced visibility and reach ✓ View applicants for a job ✓ Networking opportunities ✓ Access to a wide range of employees that fits for the position
3	**System Administrator**	✓ Manage system users ✓ User Support and Assistance ✓ Continuous improvement and system enhancement

❖ The benefit for our team

As a development team, this project will provide us with a great opportunity to enhance our skills and knowledge in web development. We will be able to gain hands-on experience in developing a featured online job portal system, which requires skills in front-end and back-end web development, database management, and more.

7. Methodology

7.1. Data Gathering Methods

Data gathering is pivotal in understanding the tasks and activities to be incorporated into the development of the online job portal. To achieve this, insights from the existing system are essential.

- **Primary sources**

Observation and interviews will be utilized to gain firsthand and accurate knowledge about the current practices in the job market. Our team will observe the operations in employment sectors and conduct interviews with employers and job seekers to understand their needs and challenges.

- **Secondary sources**

Document analysis will be implemented, involving a review of existing literature, project reports, and other documentation to gather comprehensive insights.

7.2. Methodology Development

For the analysis and design phase, the project will employ Object-Oriented Methodologies. This approach allows for the representation of complex relationships among different entities, ensuring a consistent notation throughout the system.

7.3. Developmental Approach

For our project development approach, we preferred to use the Spiral Model. We selected this model for its effective risk management and adaptability to the projects. The Spiral Model is one of the most important development Life Cycle models, which provides support for Risk Handling [1]. This model facilitates the evolution of the system through iterative enhancements, allowing for flexibility in accommodating changing requirement. The benefits of the Spiral Model, such as ease of testing and debugging during smaller iterations, make it a suitable choice for the development of the online job portal. This approach aligns with the dynamic nature of the project, ensuring that risks are identified and resolved throughout the development process.

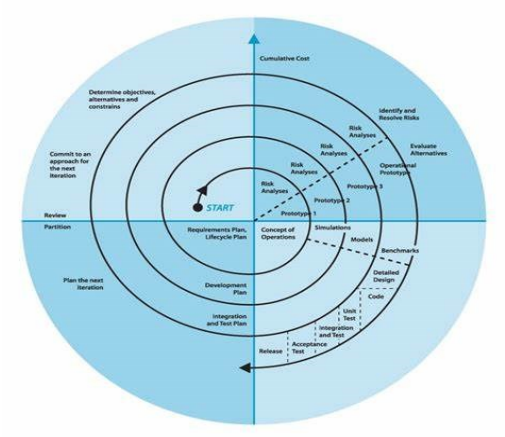

Figure 1: spiral mode

7.4. Implementation Strategy

The implementation strategy will follow a phased approach, aligning with the spiral cycles of the chosen model. This iterative implementation allows for parallel development, making the system more adaptable to changing requirements and facilitating ongoing testing and debugging.

7.5. Design Architecture

Our project will utilize a one-tier architecture during the development and testing phase, and will transition to a three-tier architecture upon completion. The one-tier architecture is chosen for its simplicity and efficiency in supporting the project's initial requirements. Once the project is finished, we will implement a three-tier architecture.

7.6. Testing and Quality Assurance

Quality assurance will be integral to each iteration, with testing conducted at various stages of development. The iterative nature of the Spiral Model allows for early identification and resolution of issues, ensuring that the system meets the appropriate outcome.

7.7. Continuous Improvement and Feedback Loop

After the initial deployment, a continuous improvement strategy will be implemented. This involves ongoing monitoring of user interactions, analyzing system performance, and collecting feedback from both employers and job seekers. Regular updates and refinements will be made based on this feedback loop, ensuring that the online job portal remains aligned with the evolving needs of the users and the dynamic job market.

7.8. Resource Management

Efficient allocation and management of resources, including personnel, time, and technology, will be a priority. This involves optimizing the development team's skills, adhering to project timelines, and ensuring that the necessary technological infrastructure is in place. Effective resource management will contribute to the successful execution of each development iteration.

8. Developmental Tools

Here are known Hardware and software tools that will be used throughout the project. In addition, we will keep an open mind and explore new technologies that might improve the efficiency and effectiveness of our project. We believe that staying up-to-date with the latest developments in the field will help us achieve better results and deliver a superior experience to our clients.

Table 1:Hardware Tools

Tools	Quantity	Description
Desktop	1	To install and check different OS
Laptop	1	To develop the project and document preparation
Mobile Phone	2	To check the compatibility of the project
External Hark Disk	2TB	To store Software's and backup the daily project file

Table 2:Software Tools and Their Purposes

Tools	Description
Microsoft Office 2019	To prepare documentation and preparation
Microsoft Edge / Firefox/ Chrome	To browse the internet
Virtual Code Studio	To Write and edit Source Code
Xampp /Wampp Server	Run PHP File on Local host To create Database
E-draw Max	To develop diagrams
Windows 10/11 OS, Linux OS, Ubuntu, Mac OS, Emulators	To run other software

9. Alternative Solutions and Proposed Solutions

9.1. Alternative Solution

We consider an Online web-based System as an alternative solution to solve the problem. The proposed alternative solution involves the development of an online web-based system to address the challenges identified in the existing job market. An online web-based system is accessible over a network connection using HTTP, providing a versatile platform for users to access and retrieve information. This alternative suggests replacing the current job search and application methods with a good one that facilitates seamless interaction between employers and job seekers.

Key Features of the Alternative Solution:

- **Accessibility**: Users can access the system from any computer connected to the internet using a standard browser plus they can also access the system from their mobile phones. Choosing a website over mobile apps or desktop software makes it easier for everyone to access the Online Job Portal System from any device. This way, everyone can benefit from it and there are no limitations.
- **Information Retrieval**: The System allows users to easily retrieve and access information relevant to job opportunities, candidates, and application processes.
- **User Permissions**: Users can access information based on permission login.

9.2. Proposed Solution

Our proposed solution is the development of an online job portal system that is called OJPS, designed to operate within a web-based environment. The system aims to provide an effective and resource-saving platform for users, catering to the specific needs of employers and job seekers. The proposed online job portal system introduces several significant improvements over traditional methods.

Key Features of the Proposed Solution:

- **Online Web-Based Operation**: The system operates within a web-based environment, offering users the flexibility to access it from any device with an internet connection.

- **Workload Reduction:** The new system is designed to reduce the workload for both employers and job seekers, streamlining the job search, recruitment, and application processes.
- **Efficiency Improvement:** By minimizing the time and effort needed for tasks such as job diagnosis and treatment, the proposed solution enhances the overall efficiency of the employment process.

Benefits of the Proposed Solution:

- **Enhanced User Experience:** The online job portal provides a user-friendly interface for a seamless experience, ensuring accessibility for all users.
- **Time and Resource Savings:** The system contributes to resource savings for both employers and job seekers by optimizing job-related processes.
- **Improved Task Performance:** The proposed solution aims to improve task performance in job-related activities, fostering a more efficient and effective employment connection platform.

Generally speaking, while the alternative solution suggests the adoption of a web-based application, the proposed solution advocates for the development of a comprehensive online job portal system. The proposed solution aligns with the specific requirements of the job market, aiming to enhance accessibility, reduce workload, and improve the efficiency of employment-related tasks for all stakeholders involved.

10. Feasibility Study

The feasibility study outlines the project's viability across various dimensions, providing a comprehensive assessment of its potential success and positive impact.

10.1. Political Feasibility

The political feasibility of the Online Job Portal System is evident in its alignment with national employment objectives and policies. As the project aims to enhance employment connections in the Ethiopian job market, we believe that it supports governmental initiatives related to job creation and economic development.

10.2. Social Feasibility

Socially, the project addresses the needs and aspirations of diverse user groups within the Ethiopian community. By providing a platform that considers users with varying educational backgrounds, the system promotes social inclusivity and equal access to job opportunities. The emphasis on user-friendly features and an intelligent assistant contributes to a positive social impact, facilitating a more user-centric and supportive job search experience.

10.3. Economic Feasibility

From an economic standpoint, the project demonstrates feasibility through its cost-free access to essential features for users. By eliminating financial barriers, the system ensures that job seekers, including those with limited financial means, can fully utilize the platform. The economic benefits extend to employers, who can access a streamlined recruitment process, potentially reducing costs associated with traditional hiring methods.

10.4. Operational Feasibility

Operationally, the Online Job Portal System is designed to streamline the recruitment process for employers and provide a user-friendly experience for both job seekers and administrators. The phased development approach allows for continuous improvement based on user feedback and emerging industry trends, ensuring operational adaptability. The integration with external platforms enhances overall functionality, contributing to the system's operational efficiency.

10.5. Technical Feasibility

From a technical perspective, the project's feasibility is supported by advancements in web development technologies. The choice of a website as the primary platform ensures compatibility with a wide range of devices, promoting accessibility. The implementation of security measures safeguards user data, addressing technical concerns related to data integrity and confidentiality. Technical feasibility is further enhanced by the project's adaptability to emerging industry standards and technological advancements.

11. Schedule of activities and Budget breakdown

Table 3: Schedule of activities

Phase	Tasks	Begin Date	End Date	Duration
1st	Title Selection	October 5/2023	November 5/2023	30 days
2nd	Proposal Preparation	November 11/2023	November 17/2023	6 days
3rd	Requirement elicitation	November/18/2023	November/30/2023	12 days
4th	Software Requirement specification	November/30/2023	December/7/2023	7 days
5th	Software design	December/7/2023	January/7/2024	30 days
6th	Implementation	January/10/2024	May/18/2024	130 days
7th	Unit and system testing	May/19/2024	June/1/2024	10 days

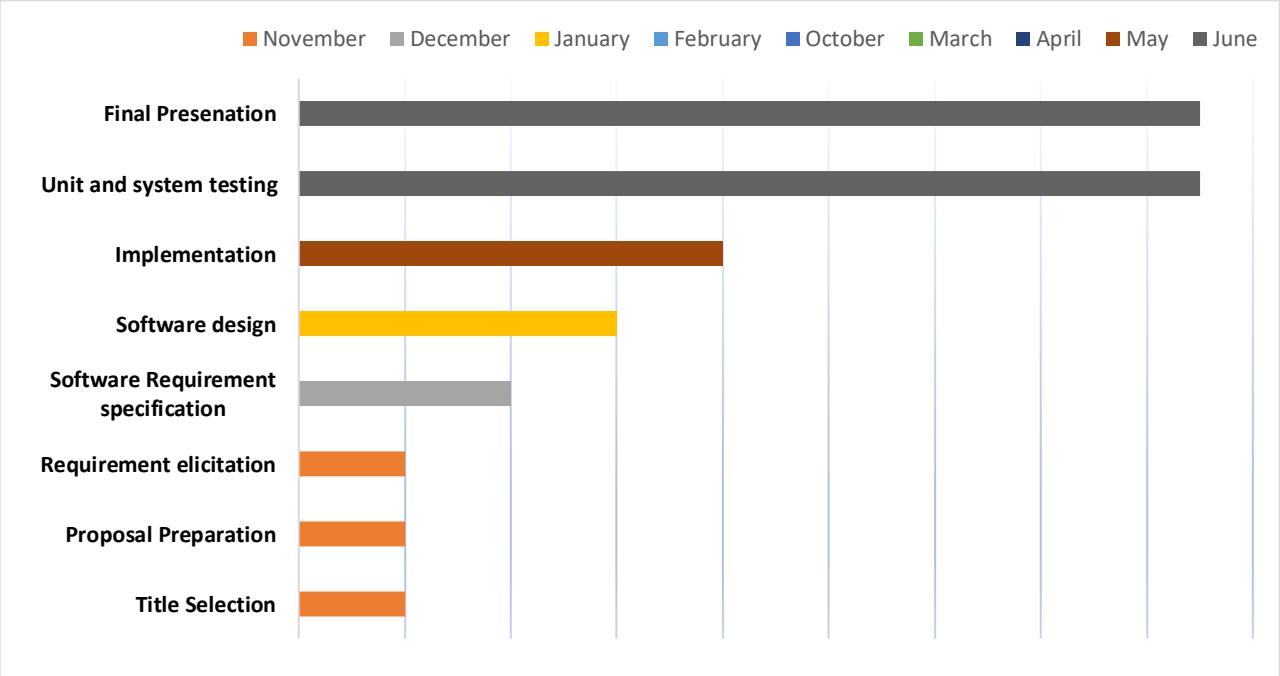

Figure 2: Schedule of Activities chart

Table 4: Budget Breakdown

Hardware Tools and others			
Items	Quantity	Per unit cost	Total cost
1, Desktop	1	25,000 ETB	25,000 ETB
2. Laptop	1	70,000 ETB	70,000 ETB
3. Hard disk	1	4,000ETB	4,000 ETB
4. Internet Connection	unlimited	6,000ETB	6,000ETB
Total			105.000ETB
Software's			
Names	**Quantity**	**Per cost**	**Total cost**
1, windows 10 OS	1	300.00ETB	300.00ETB
2, windows 11 OS	1	500.00ETB	500.00ETB
3, Linux OS	1	free	free
4, Mac OS	1	free	free
5, Xampp	1	free	free
6, Visual Studio Code	1	free	free
Total			800.00ETB
Stationary Materials			
Items	**Quantity**	**Per unit cost**	**Total Cost**
1, Paper	4 dozen	800 ETB	3,200ETB
2, Pen	2 dozen	1000 ETB	2,000ETB
3, pencil	1 dozen	356ETB	356ETB
4, Binder	1 Piece	375ETB	375ETB
Total			5,200ETB
Total Project Estimated Cost			111,731ETB

Summary

The proposed project aims to address the evolving needs of the Ethiopian job market by developing an advanced online job portal system. Recognizing the limitations of traditional job-seeking methods, the project seeks to create a user-friendly platform accessible to a wide range of individuals, bridging the gap between job seekers and employers. The system will offer distinct features for both employers and job seekers, including job listings, personalized profiles, and a streamlined application process.

The project's objectives include gathering and analyzing requirements, designing the system using Object-Oriented Methodologies, implementing it based on the design, testing for functionality, and deploying it in a working environment. The development will follow a phased approach, allowing for continuous improvement based on user feedback. The proposed system is expected to have a positive impact on the academic and professional development of the project team, enhancing their skills and knowledge in web development.

The feasibility study indicates political, social, economic, operational, and technical feasibility, emphasizing the project's alignment with national employment objectives and policies, social inclusivity, cost-free access for users, operational adaptability, and compatibility with emerging web development technologies.

The schedule of activities outlines the project timeline, starting with title selection and concluding with unit and system testing. A budget breakdown includes estimated costs for hardware, software, and stationary materials, ensuring efficient resource management

In conclusion, the proposed online job portal system addresses the challenges faced by job seekers and employers in Ethiopia, offering a comprehensive and efficient solution. The project's structured approach, feasibility considerations, and budget breakdown contribute to its potential success in revolutionizing the Ethiopian job market.

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